

Operator Discovery Report: rank_2opt_candidates_cluster_span_balanced

- Operator type: candidate_ranker.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic candidate ranker for 2-opt style local moves: prioritizes moves that reduce estimated crossings/long edges while rewarding productive span and mild cluster alignment; includes trap and nearest-neighbor pena
- Novelty classification: trap_aware_ranker.
- Rediscovery signature: candidate_ranker:delta=1.250,cross=1.000,span=0.400,trap=0.600.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- For each candidate move, compute a composite score from immediate delta, crossing removal, span, trap awareness, and local structure.
- Sort candidate moves by ascending composite score.
- Apply the first move that yields a genuine tour improvement.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: fe95dee0aba78001.

Why It Should Help

- The operator changes which local-search moves are tried first, so it can improve search efficiency without rewriting the underlying solver.

Failure Cases

- Underperforms on grid_like_tsp with gap delta 0.054436.
- Does not transplant cleanly into nearest_neighbor_2opt.
- Does not transplant cleanly into random_restart_2opt.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.003328, runtime delta 41.119457, distance-eval delta -100.571429.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 1476.261843, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.006029, runtime delta 38.963829, distance-eval delta 769.714286.
- sparse_three_opt gap delta 0.0, runtime delta 2.402457, distance-eval delta 0.0.

- clustered_local_search gap delta -0.002014, runtime delta 31.334871, distance-eval delta 1922.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.238386 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.161132 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.006956.
- Runtime inflation: 0.095546.
- Distance-evaluation inflation: 0.003718.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.006956.
- simplified_delta_ranker mean gap 0.127067 and delta vs full operator -0.001206.
- shuffled_ranker mean gap 0.128189 and delta vs full operator -8.4e-05.